

FEMSeg_kaz_v2: Short Execution Guide (100k Wordforms)

This document provides a concise, step-by-step guide for training and testing the FEMSeg_kaz_v2 neural morphological segmenter on a subset of 100,000 Kazakh wordforms.

1. Environment Setup

The experiments can be run either in Google Colab or on a local machine with Python 3.9 or higher.

Required dependency:

- pytorch-crf

2. Data Preparation

Input corpus:

merge_seg_wordforms.txt (approximately 2.33 million CSE-segmented Kazakh wordforms).

From this corpus, 100,000 wordforms are randomly sampled to form a lightweight training subset.

3. Dataset Construction

Each wordform is converted from CSE notation into:

- a character sequence,
- BMES labels (Begin, Middle, End, Single).

The dataset is split into:

- 90% training set,
- 10% validation set.

A character vocabulary (char2id) is automatically built.

4. Model Architecture (FEMSeg_kaz_v2)

The model follows a neural sequence-labeling architecture:

- Character embedding layer,
- Bidirectional LSTM encoder,
- Linear projection layer,

- Conditional Random Field (CRF) for BMES decoding.

5. Training Procedure

- Number of epochs: 3
- Optimizer: AdamW
- Loss function: negative log-likelihood from CRF
- Evaluation metric: validation token accuracy (val_token_acc)

The best-performing checkpoint is saved automatically.

6. Inference and Testing

The trained model can be used to segment:

- individual words,
- full sentences.

Output formats:

- CSE-style segmentation,
- SentencePiece-compatible (SP) format.

7. External Evaluation

Segmentation quality is evaluated using a small manually annotated gold set (10 sentences), computing:

- Average normalized edit distance,
- Boundary-based metrics (TP, FP, FN, TN),
- Accuracy, Precision, Recall, and F1-score.

8. Practical Outcome

The resulting FEMSeg_kaz_v2 model:

- generalizes to unseen wordforms,
- produces linguistically plausible morpheme boundaries,
- can be integrated as a morphology-aware pre-tokenization step before BPE or mT5 models.